Assignment – II

Database Management System

(24CSH-298)

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Submitted To:

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Ans 1:- (i) Legal and ethical issues:

Legal and ethical issues relate to laws, regulations and moral responsibilities in handling database information.

- Privacy Protection: Personal or scientific data (employee details, salary, medical data) must be protected.
- Data Misuse: Unauthorised use or sharing of confidential information is unethical and illegal.
- Intellectual Property Rights: Database may contain copyrighted information.

Example: Leaking a customer personal information or using employee data without consent.

(ii) Policy issues: These relate to the rules and guidelines established by an organization to protect its database systems. Organizations develop security policies that define how data should be accessed, used and protected. These policies specify user roles, access rights.

(iii) System Related Issues: System related issues arise from technical weakness in database system infrastructure, like weak authentication issues, lack of updates etc.

Eg:- A hacker exploiting a database vulnerability to steal information.

Ans 2:- (i) Access control:

- It restricts who can access the database and what operations they can perform, where one given specific ~~information~~ permission (read, write, update).
- Implemented using user account roles and privileges.
- E.g: A clerk can view records but cannot modify them.

(ii) Inference Control:

- Inference control prevents users from deducing sensitive information from accessible data.

E.g: Preventing someone from calculating an individual's using average salary queries.

(iii) Flow control:

- Flow control ensures that information flow from higher security levels to lower security levels.
- It restricts how information moves within a system.

E.g:- Preventing secret data from being transferred to public database.

(iv) Data Encryption:

- It converts plain text data into coded from cipher text to protect it from unauthorised access.

E.g:- Encrypting credit card details stored in a database.

Ans 3:- A view is a virtual table created from one or more tables. It stores a query rather than actual data.

SQL Query:

```
CREATE VIEW Employee View As
SELECT Emp ID, Emp Name, Dept ID
FROM Employee ;
```

This view will display only Emp ID, Emp Name, and Dept ID hiding other columns such as salary.

Problem List | Submit | 0 | Premium

Description | Accepted | Editorial | Solutions | Submissions

All Submissions

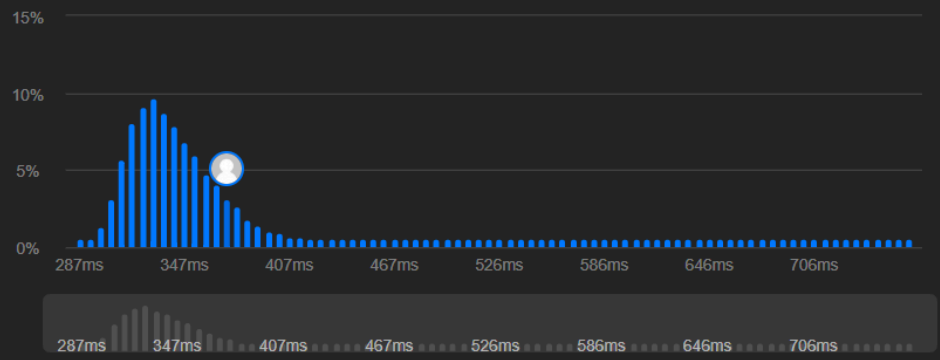
Accepted 11 / 11 testcases passed

AnshRana02 submitted at Mar 16, 2026 16:06

Solution

Runtime

371 ms | Beats 23.46%



Code | MySQL

```
SELECT e1.name
FROM Employee e1
JOIN Employee e2
ON e1.id = e2.managerId
GROUP BY e1.id, e1.name
HAVING COUNT(e2.managerId) >= 5;
```

Code

MySQL | Auto

```
SELECT e1.name
FROM Employee e1
JOIN Employee e2
ON e1.id = e2.managerId
GROUP BY e1.id, e1.name
HAVING COUNT(e2.managerId) >= 5;
```

Saved | Ln 6, Col 33

Testcase | Test Result

Accepted Runtime: 80 ms

Case 1

Input

Employee =

Proble... Submit Premium

Description | Accepted | Editorial | Solutions | Submissions

Submit Ctrl Enter

le

All Submissions

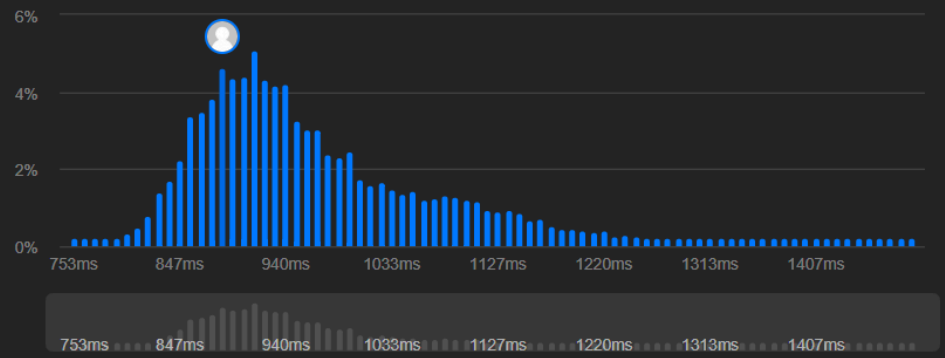
Accepted 21 / 21 testcases passed

AnshRana02 submitted at Mar 16, 2026 16:07

Solution

Runtime

892 ms | Beats 76.81%



Code | MySQL

```
SELECT e.name, b.bonus
FROM Employee e
LEFT JOIN Bonus b
ON e.empId = b.empId
WHERE b.bonus < 1000 OR b.bonus IS NULL;
```

MySQL Auto

```
SELECT e.name, b.bonus
FROM Employee e
LEFT JOIN Bonus b
ON e.empId = b.empId
WHERE b.bonus < 1000 OR b.bonus IS NULL;
```

Saved Ln 5, Col 41

Testcase | Test Result

Accepted Runtime: 94 ms

Case 1

Input

Employee =